

Intelligenza Artificiale

“Deep learning (part 4)”

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Lecture outline

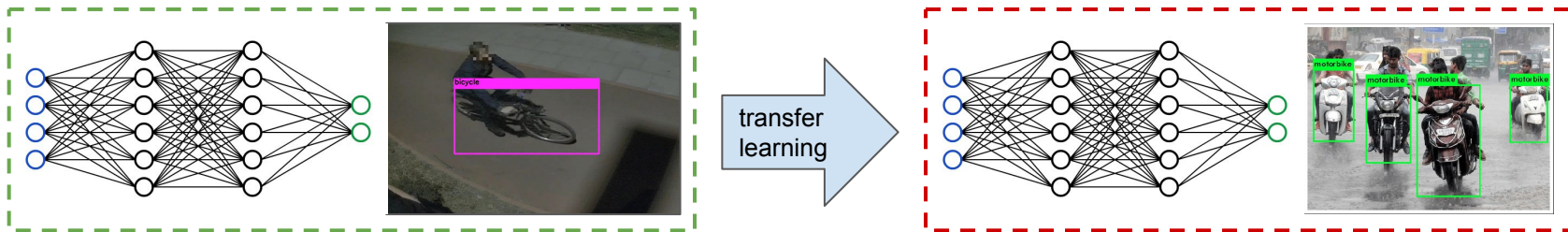
- Transfer learning
- Language models
- Attention
- Transformers



Slides partly based on Russell & Norvig instructors material <http://aima.cs.berkeley.edu/instructors.html> and on Jurafsky & Martin slides <https://web.stanford.edu/~jurafsky/slp3/>

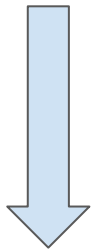
Transfer learning

- With **transfer learning**, experience with one task helps an agent learn better on another task
- General approach for neural networks:
 1. first copy over the weights learned for **task A** to a network that will be trained for **task B**
 2. then update the weights as usual (e.g. with gradient descent) using data for **task B**



Examples of transfer learning

RoBERTa (base) language model

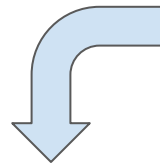


Medical or financial expert

Autonomous car in simulation

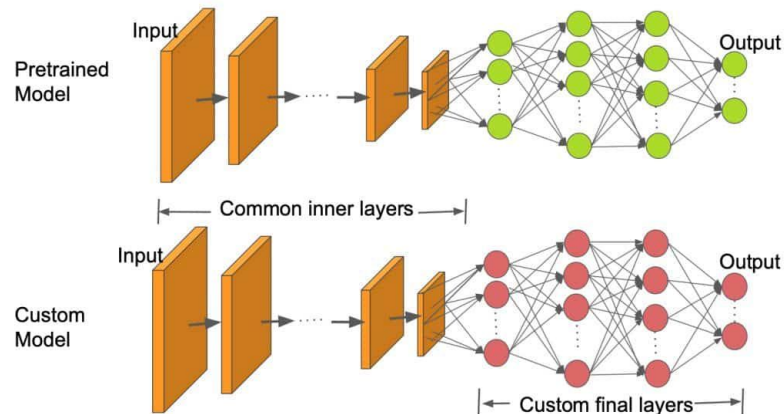


Real autonomous car



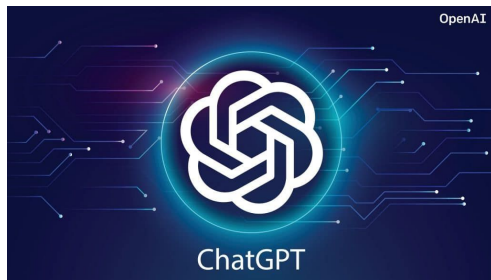
Transfer learning in practice

- Practical steps
 1. Freeze the first few layers of the pretrained model that serve as feature detector
 2. Modify the parameters of the higher levels only
 - problem-specific features and classification
- Common to start with popular pretrained model (e.g. ResNet-50 for object recognition)
- Then **fine-tune** the model in two ways:
 - giving examples in the desired domain
 - training on the task to be performed



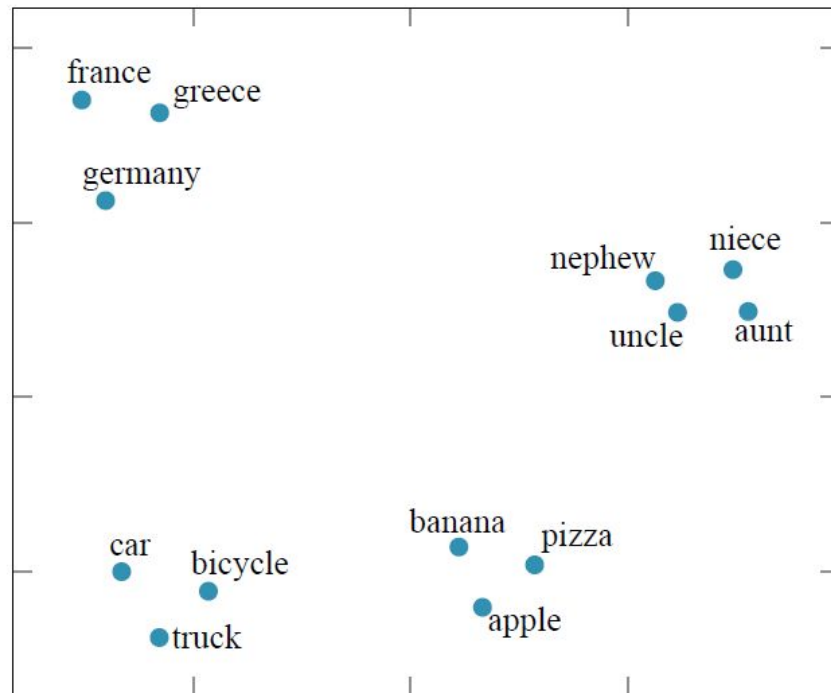
Language models

- Language models are (probabilistic) models of natural language
- **Large language models (LLMs)** are artificial neural networks that can achieve general-purpose language understanding and generation
- A form of “generative AI” – take an input text and repeatedly predict the next word
- Their use extends beyond language (e.g. LVM – Large Vision Model)



Word embedding

- Word, or part of it, represented by a number (**token**)
 - also spaces and punctuation
- **Word encoding** converts words into numbers, without caring about neighbours
- The whole set of tokens of a LLM constitutes its **vocabulary** (can be huge!)
 - E.g. GPT-2's vocabulary has ~50k tokens
- **Sentences** are sequences of tokens
- **Word embedding** goes beyond encoding by clustering related words in a lower dimensional space of tokens



Token prediction



- Given a sequence of tokens (i.e. **prompt**), the LLM predicts the probabilities of the next tokens
 - i.e. given ['To', ' be', ' or', ' not', ' to', ' be'] GPT-2 predicts the ~50k probabilities of all its vocabulary tokens
- We could then use a **greedy selection** – i.e. highest probability – or a **random selection** based on the probability distribution of the tokens
 - Random approach introduces more variety
 - *Greediness* can be tuned to be more or less conservative in the selection

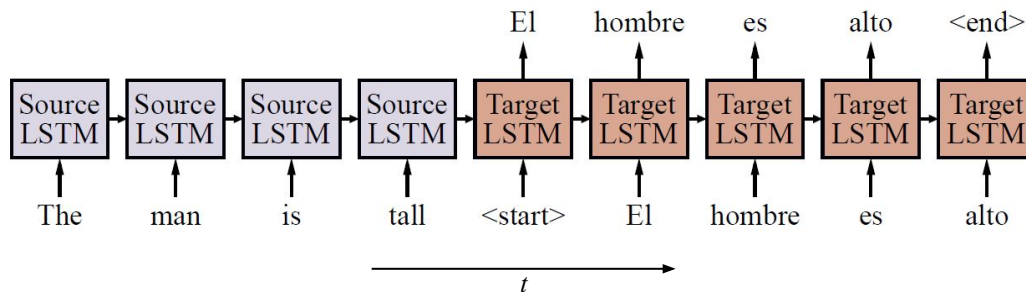
Text generation

- Whole sentences can be generated by looping on the prediction
 1. **sentence** = **prompt**
 2. **token** = predict from **sentence**
 3. **sentence** = append **token** to **sentence**
 4. if **token** == **period** '.'
 - **END**
 5. go to step 2



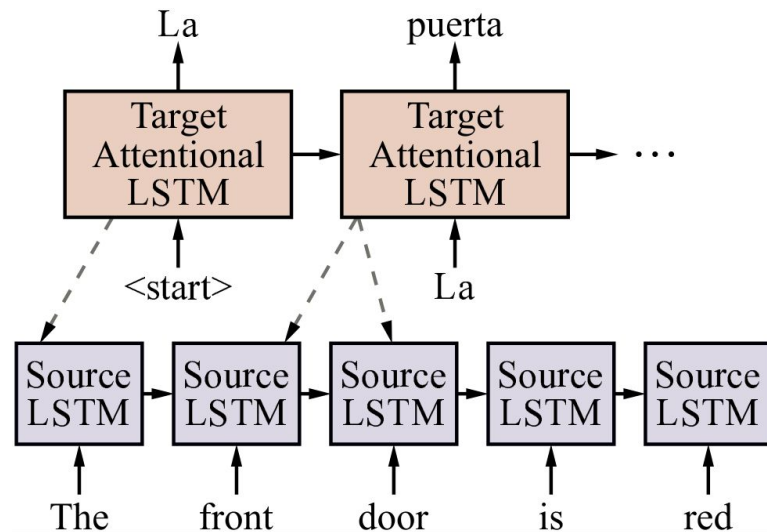
Sequence-to-sequence models

- How to predict tokens?
- One way is to use a **sequence-to-sequence model**
 - E.g. using two LSTM-based RNNs, one for the **source** and one for the **target**
 - The source RNN maintains some information about neighbouring tokens
 - This information, related to the context of the words, is transmitted to the target RNN for predicting new sequences
- E.g. can be applied to machine translation (here a hidden layer)



Attention

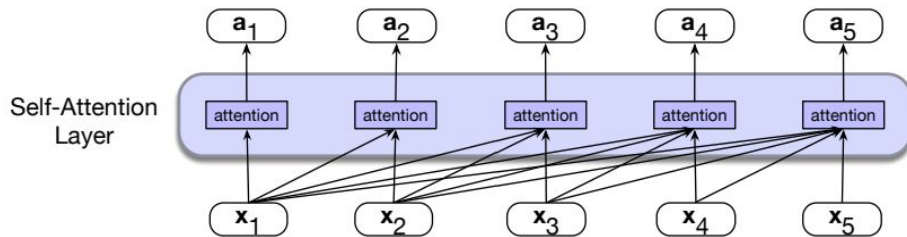
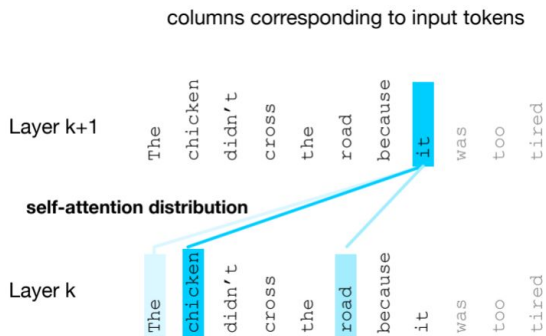
- **Problem:** RNNs tend to forget old information e give more importance to the most recent tokens (even with LTSMs)
 - remember the Markov property of RNNs...
- What if we consider ALL the source units?
 - computationally not feasible
- But we can introduce some **attention** units that selects a small part of it, which better captures the **context** of the words
 - applied to both source and target
 - latent, the model learns what to use



Self-attention



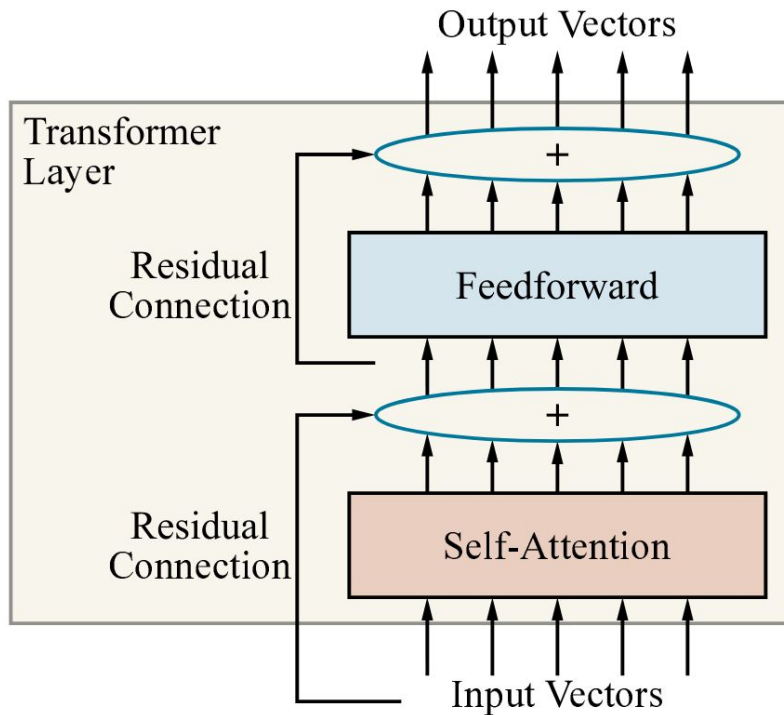
- **Self-attention** – hidden states attend themselves
 - this better captures nearby and long-distance context



- Not easy to implement though, because hidden recurrent layers would be naturally biased towards attending themselves (similar to previous RNN without attention)

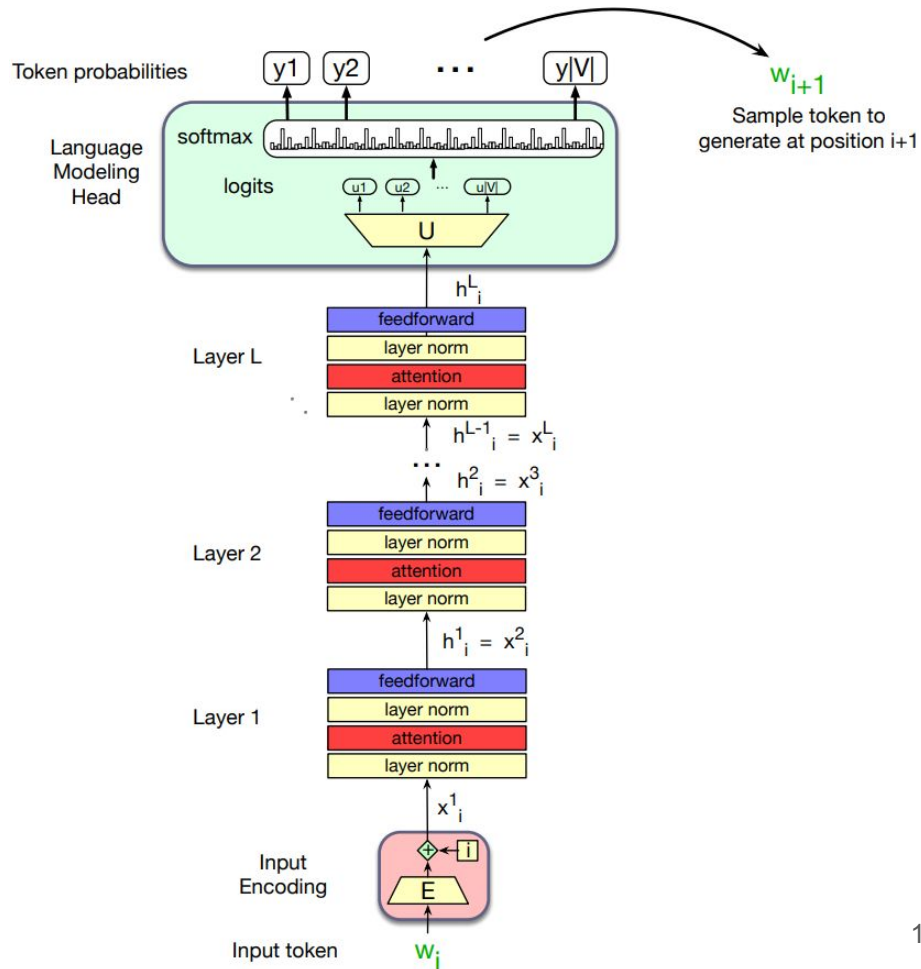
Transformers

- **Transformers** implement **self-attention**
- They are at the core of LLMs such as **GPT (Generative Pre-trained Transformer)**
- Each transformer layer consists of several sub-layers
- Transformer (encoder) includes
 - self-attention layer
 - feedforward layers
 - residual connections



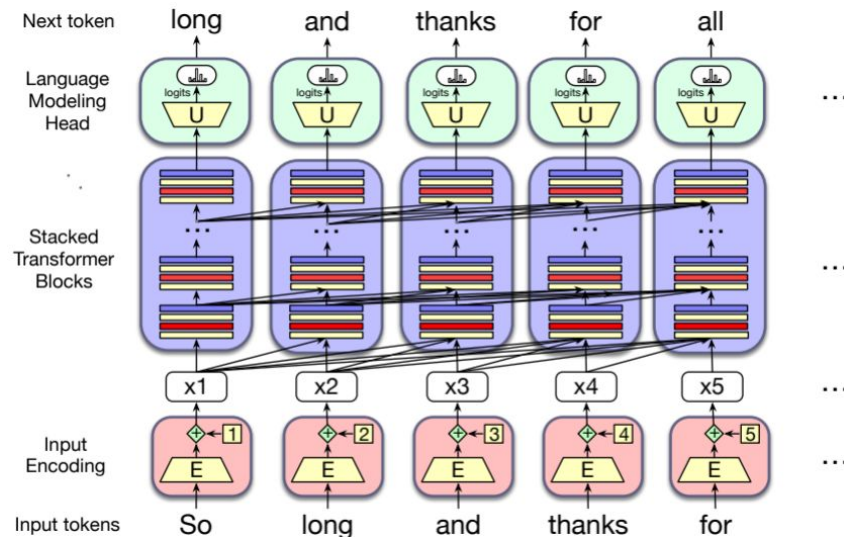
Transformers

- Stack of sub-networks (blocks)
 - input token at the bottom
 - token probabilities at the top
- Three major components
 - **Input encoding** block to process tokens
 - Column of transformer blocks
 - each one is a multi-layer network
 - typically 12 to 96 blocks
 - **Language model head** to produce token probabilities



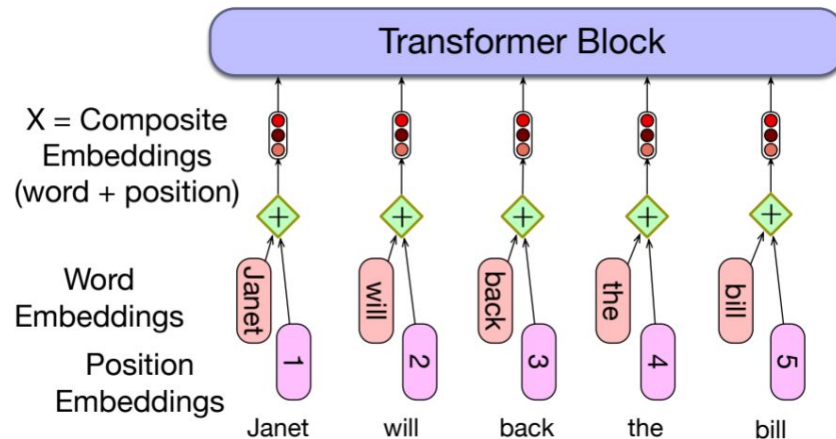
Transformers

- Parallel transformer “columns” are connected by attention links
- The number of columns corresponds to the number of input tokens and represents the **context width** (i.e. information breadth)
 - ~128k for modern LLMs
- The vertical number of blocks in each transformer determines the **context depth** (i.e. abstraction level)
 - e.g. the token “cut” could be
 1. word or acronym
 2. noun or verb (present or past tense?)
 3. related to clothes, humans, movies
 4. ...

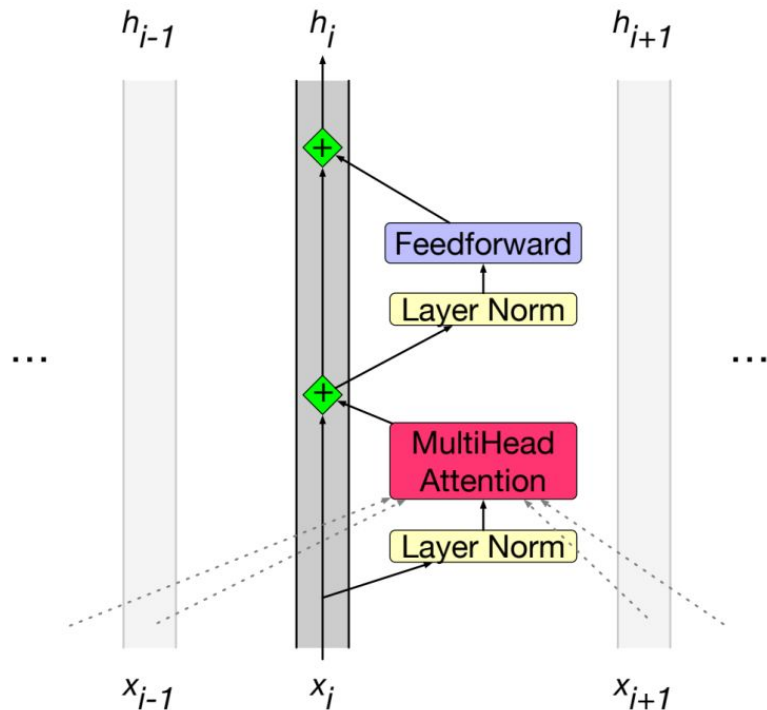


Input encoding

- Two elements are merged
 1. Embedding of the i -th **word**
 2. Embedding of the **position** i
- Possible problem is that, during training, there are less (very) long sentences
 - This might not generalise very well
- Alternative is to use **relative position**, implemented directly in the attention layers rather than the input



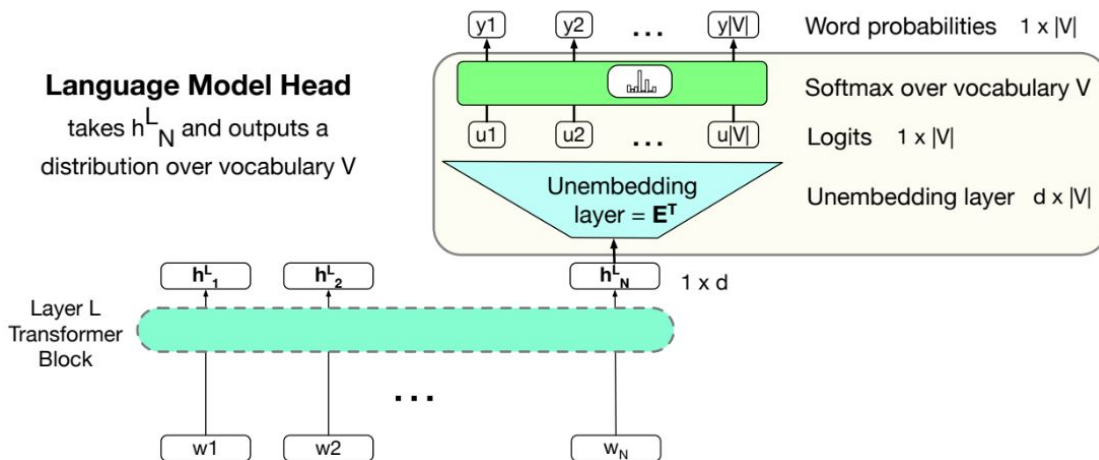
Residual stream



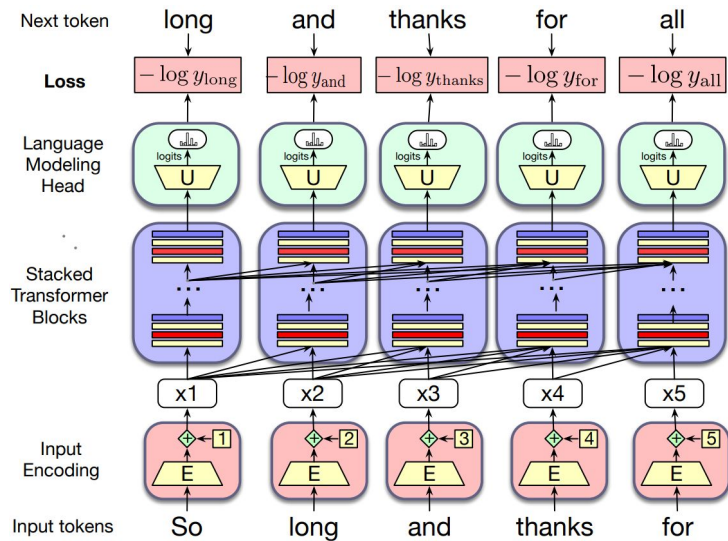
- Common representation of the transformer block is the *residual stream*
- Each block has four elements
 1. **LayerNorm** to normalise the input
 - to avoid vanishing/exploding gradients
 2. **MultiHead Attention** to attend also to the other inputs
 3. Another **LayerNorm**
 4. **Feedforward**, which is a fully-connected 2-layer network (1 hidden + 1 output)
 - usually size of hidden layer larger than input size

Output

- Maps from the output of the last transformer's block to a probability distribution of words from a vocabulary $\mathbf{V}$
- The unembedding is effectively the transpose $\mathbf{E}^T$ of the input embedding
- A *softmax* layer transforms the unembedded tokens into probabilities
 - The word can be chosen by sampling from these probabilities



Training



$$\dots = \frac{1}{T} \sum_{t=1}^T L_{CE}$$

- LLMs are trained with **cross-entropy loss** (more precisely *negative log loss*) and gradient descent
- During training, the probability assigned to the correct word is used to calculate the cross-entropy loss
- The loss is averaged over the entire sequence of words

Limitations

Output of GTP-4o (June'24)



explain this image



This image is a humorous cartoon that uses a visual gag to play on social conventions. It shows three people walking: a young woman in a revealing outfit, a little boy holding her hand, and an older woman with an exaggeratedly shocked expression looking back. The young woman and the boy are walking away, while an older man, walking beside the older woman, seems oblivious to what is happening.

The punchline lies in the old lady's reaction. Her shocked expression and the caption "Why did the old lady look back?" suggest that she is reacting to the boy, who appears to be looking at the young woman's behind. The humor arises from the innocence of the boy juxtaposed with the suggestiveness of the situation, and the old lady's exaggerated reaction adds to the comedic effect.



Output of Gemini 3 (December'25)

This image is a classic visual joke based on a forced perspective optical illusion.

The Optical Illusion

At first glance, it looks like the small boy walking on the left has an impossibly long, pointed nose that is touching the woman in front of him. This creates a suggestive and scandalous-looking scene.

The Reality

The "nose" is actually not part of the boy at all. It is the arm and hand of the tall man on the far right. Because the characters are lined up perfectly, the man's arm overlaps with the boy's face, creating the illusion of a long nose.

Why the lady looked back

The text asks, "Why did the old lady Looked back?" (grammar error aside). The answer is that from her perspective, she sees the same scandalous illusion that the viewer sees initially. She is looking back in shock and judgment at what appears to be a very strange and inappropriate situation.

Conclusions

- Suggested reading
 - Ch. 22, sec. 22.7 of Russell & Norvig
 - Ch. 25, sec. 25.1-25.4 of Russell & Norvig
 - Ch. 8 of Jurafsky & Martin (online)
<https://web.stanford.edu/~jurafsky/slp3/>
- Last lecture!
- Questions?

